

INTRODUCTION

Hello, I am WenHao. I graduated from **Nanyang Technological University (NTU)** with a **Bachelor's degree in Mechanical Engineering, specialising in Robotics and Mechatronics**. I am a logical and adaptable person with a strong background in Math, fueled by curiosity and is always up for a challenge. Curiosity fuels my desire to tackle intricate challenges and contribute insights in a dynamic, collaborative space. I am a highly motivated learner and seeks for an opportunity to expand my knowledge base in the field of Robotics.

EDUCATION

Nanyang Technological University, Singapore **Bachelor of Mechanical Engineering**

Aug 2022 – May 2026

- Specialisation - Robotics and Mechatronics
- Honours (Distinction), CGPA 4.10

Relevant Coursework – Introduction to Computational Thinking (Python), Introduction to Data Science and Artificial Intelligence, Introduction to Electrical Circuits and Electronic Devices, Mechatronics System Interfacing, Introduction to Mechatronics Systems Design, Robotics, Mechatronics Engineering Design.

Southern University of Science and Technology, Shenzhen, China (Exchange)

Feb 2025 – June 2025

Relevant Coursework – Mechanical Design, Control Theory, Measurements and Sensing systems.

SKILLS

- Computer skills: Python, C++, Solidworks, Fusion360, Microsoft Office, Excel, Word, PowerPoint, MATLAB
- Language: English, Chinese, Basic Cantonese

ACADEMIC PROJECTS/COMPETITIONS

Final Year Project with Humanoid Robot (MA4079) (NTU)

Sep 2025 – May 2026

- Developed a teleoperation system for a Unitree G1 humanoid robot using IMU sensors (ReboCap) via Ngrok.
- Implemented and evaluated reinforcement learning algorithms using Isaac Gym.
- Designed an IMU-based motion imitation learning pipeline using ReboCap within the MjLabs framework.
- Built a video-to-motion imitation pipeline leveraging GVHMR and GMR frameworks.
- Conducted comparative analysis of both pipelines in terms of stability and accuracy.

Mechatronics Engineering Design (MA4012) (NTU)

Feb 2026 – April 2026

- Collaborated in a 9-member team to design and build an autonomous robot for tennis ball collection within a defined zone.
- Contributed to the mechanical design and fabrication of the robot's chassis and drivetrain system.

Robotics (MA4825) (NTU)

Aug 2025 – Nov 2025

- Picked up concepts on forward and inverse kinematics and learned the ABBRoboStudio software.
- Developed a pick-and-place simulation using ABBRobotStudio with an IRB1300 robot arm and RobotWare for OmniCore controller.
- Workpieces and environment components are modelled in SolidWorks and integrated into the simulation environment.

MaritimeONE Case Summit 2025

Aug 2025 – Sep 2025

- Collaborated in a team of 4 to analyse the Panamax dry bulk shipbroking landscape, identifying emerging market risks and regulatory challenges.
- Conceptualised innovative, practical AI strategies and market intelligence tools to optimise workflows and decision-making for shipbrokers.

Measurements and Sensing Systems (SUSTech)

Feb 2025 – June 2025

- Designed a real-time spine visualisation with IMUs and Flex sensors.

- Made use of MATLAB and Arduino for the software components.

DSTA BrainHack TIL-AI (Today I Learned - Artificial Intelligence)

May 2024 – June 2024

- Worked in a team to help clean datasets, track model testing, and manage project tasks.

Mechatronics Systems Interfacing (MA2011) (NTU)

Jan 2024 - May 2024

- Tasked to create a digital twin of new project in a group of 4, making use of any applications available through self-learning.
- Drafted a digital twin on a smart carpark, using Arduino as the physical representation and Unity to design the 3D model.
- Managed to setup a bi - directional communication with Unity and Arduino by opening serial communication on Unity C# script.

Engineering Innovation and Design (E.I.D) (NTU)

Jan 2024 - May 2024

- Discussed and brainstormed with a team of 7 members to research and compile relevant information for the creation of a retractable laundry pole.
- Understood one another's strengths, and split into smaller groups, overcoming the project's limitations of time, quality, and budget.

RPA and GEN AI Hackathon 2024 with SimplifyNext

Jan 2024 – March 2024

- Introduced and understood use cases to RPA, UiPath and GEN AI.
- Obtained a diploma of completion for "Introduction to Automation" by UiPath.

Introduction to Mechatronics Systems Design (MA2012) (NTU)

Aug 2023 - Nov 2023

- Designed a security system for a maze structure in a group of 4 based on the theme "The Secured Place".
- Managed to achieve 3rd runner up for the project, incorporating gears learnt such as DC motors and Servo motors into a maze structure using Arduino.

WORK EXPERIENCE

Aidrippers Singapore Pte Ltd

Jun 2024 - Dec 2024

Perception Engineer Internship

- Utilized object detection models to identify Prime Movers for the **Swap Battery Project**.
- Developed a **Camera Driver and Calibration package** in ROS1 Melodic (C++) for:
 - Dynamic adjustment of IP and USB camera parameters based on weather or time of day.
 - Image saving, rectification, and frame manipulation (rotation/flipping)
 - Chessboard-based camera calibration to obtain intrinsic and extrinsic parameters.
 - Seamless integration of driver, calibration, and rectification using a bash script.
- Designed **real-time image stitching** with OpenCV in ROS1 Melodic (C++) using feature matching.
- Enhanced past repository functionality to support live and **ROS bag-based image projection** for smoother stitching.
- Tested lightweight detection models for **ADAS (Advanced Driver Assistance System)** to detect driver behaviors like eating, sleeping, and smoking.
- Benchmarked and compared **YOLOv8** (Python virtual environment) and **YOLOv11** (Docker) performance on Jetson Nano (Python 3.6.9).
- Built two GUIs (**Tkinter/PyQT**) to visualize models and customize detection settings and classes.

Inland Revenue Authority of Singapore (IRAS)

Apr 2022 - Jul 2022

Call Center Specialist

- Learned about income tax and assisted taxpayers to file.
- Honed communication skills through customer service and assisted over 15 taxpayer's queries daily.

Flextronics Manufacturing Pte Ltd

Nov 2021 - Feb 2022

Assembly Technician

- Assembled electronic parts together in a team of 5 and managed to achieve the daily required KPI through effective teamwork.

- Orchestrated production line by diligently overseeing its operational state and rigorously ensuring the adherence to stringent quality and safety benchmarks.
- Facilitated loading and unloading of merchandise.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

MAE Robotics Club Sub-Committee

Aug 2023 - Dec 2024

- Managed the External Affairs Community of Robotics Club (EACO)
- Designed a website in a team of 3 for Robotics Club on Wix.
- Reached out and coordinated with 2 external companies to organise career talk for the Club.

Freshmen Orientation Camp Group Leader (Hall)

Sep 2022 - Oct 2023

- Facilitate social connections and cohesion through engaging activities for a diverse group of 30 freshmen.
- Ensured that programs run smoothly, and lunch catered arrive promptly.

Freshmen Orientation Camp Group Leader (Faculty)

Oct 2022 - Aug 2023

- Planning of interactive events and activities to encourage bonding throughout camp.
- Introduced freshmen to different facilities of MAE.

National Service (NS)

Jan 2020 - Nov 2021

- Served as a BMTC Commander
- Monitored and documented activities of each recruit, ensuring consistent participation in progressive trainings.

HOBBIES AND INTEREST

Travelling

Sports